

Sakila DVD Rental Analytics Project – MECE Breakdown

1. Project Overview

The objective of this project is to analyse the Sakila DVD Rental Store database and create an interactive Power BI dashboard to generate business insights and support data-driven decision-making. The analysis focuses on sales performance, customer behaviour, film inventory, and store and staff performance using Excel, SQL, and Power BI.

The final outcome is a structured dashboard that helps understand business trends and improve operational efficiency.

2. MECE Framework Structure

The project follows a MECE (Mutually Exclusive, Collectively Exhaustive) approach to ensure a structured and logical workflow.

Steps Covered

1. Data Cleaning in Excel
 2. Data Cleaning and Preparation in SQL
 3. Exploratory Data Analysis (EDA) in Excel
 4. SQL Business Analysis
 5. Power BI Data Modelling
 6. Power BI Dashboard Development
 7. Business Insights and Recommendations
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3. Data Cleaning in Excel

Activities

- Checked missing values and duplicates
- Standardized column formats
- Converted date fields into proper format
- Verified primary keys and data consistency
- Cleaned customer, rental, payment, and film tables

Outcome

Clean and structured dataset ready for SQL analysis.

4. Data Cleaning and Preparation in SQL

Activities

- Explored tables using SELECT queries
 - Checked null values and relationships
 - Joined tables using primary and foreign keys
 - Prepared structured dataset for analysis
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5. Exploratory Data Analysis (EDA) in Excel

Activities

- Created pivot tables for revenue analysis
- Identified top customers and films
- Analysed rental and revenue trends
- Built summary charts

Outcome

Understanding of revenue patterns and customer behaviour.

6. Power BI Data Modelling

Activities

- Connected payment, rental, customer, film, inventory, store, and staff tables
- Created relationships and DAX measures
- Built structured data model

Outcome

Ready-to-use model for dashboard creation.

7. Power BI Dashboard Development

Executive Overview

- Total Revenue
- Total Rentals
- Total Customers
- Revenue by Country and Category

Sales Analysis

- Monthly Revenue Trend
- Top Films by Revenue

Customer Analysis

- Customer Segmentation
- Revenue Distribution

Film and Inventory Analysis

- Film Category Distribution
- Rental Duration Distribution

Store and Staff Performance

- Revenue by Store
- Rentals by Staff

8. Key Insights and Recommendations

Insights

- Revenue varies across months
- Few films generate most revenue
- High-value customers contribute more
- Store and staff performance differs

Recommendations

- Increase inventory of popular films
- Focus on high-value customers
- Monitor staff performance
- Use Power BI dashboard for decision-making